



Maritime &
Coastguard
Agency

Guidance

MGN 652 (M+F) Amendment 1 infectious disease at sea

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Summary

This guidance note provides information and recommendations on

- the causes and prevention of infectious diseases on board ship
- appropriate immunisations against infectious diseases likely to be encountered by seafarers in the course of their duties
- specific requirements for prevention of insect borne diseases including malaria
- where to get further advice

Annexes are attached as follows:

Annex 1: Useful references for further information

Annex 2: UK routine and job specific immunisations and geographical requirements

Annex 3: Insect borne diseases including malaria

Amendment 1 includes no substantive changes, other than removal of specific links for www.gov.uk (<http://www.gov.uk>) and www.nhs.uk (<http://www.nhs.uk>) for COVID-19 information. Some minor editing has been carried out for greater clarity.

1. Background

1.1 Infections can cause life-threatening health problems for seafarers and lead to major costs and disruption for maritime employers. Preventive measures are therefore essential.

1.2 Infection may be transmitted by many means including by food and water, from person to person or by insect or animal bites.

1.3 The major means of infection control on board ship are through effective personal hygiene measures and work practices designed to minimize these infection risks. Such practices are not the subject of this MGN. Further information can be found in the Code of Safe Working Practices for Merchant Seafarers and Wellbeing at Sea: A Pocket Guide for Seafarers (see annex 1).

1.4 Effective hygiene must be supplemented by specific preventive measures including: - Immunisations for a range of infectious diseases (see annex 2) - Insect bite avoidance (see annex 3) - Preventative medication in the case of malaria (see annex 3).

1.5 Treatment after suspected exposure to an infection or on suspicion of signs suggestive of infection is recommended for a few conditions such as malaria (see annex 3). Post exposure treatment may also be recommended if there is a risk of HIV or rabies infection, following tick bite in high-risk areas and for some other rare bacterial and viral illnesses. Such treatment depends on an expert clinical risk assessment and is not considered here. Reference should be made to the Ship Captain's Medical Guide and telemedical advice sought.

2. Infection risks

2.1 The risk of contracting an infectious disease will depend on a number of factors. Hence requirements are determined by assessing the risk posed by:

- the route of the vessel, especially the location of ports visited, length of stay and likelihood of exposure ashore either on shore leave or when joining or leaving a vessel

- known health risks in the country of origin of seafarers and countries passed through on the way to joining or leaving the ship
- the role of each seafarer
- hygienic and clinical preventive measures taken.

3. Route and ports

3.1 It is recommended that employers and seafarers consult either their occupational health (OH) provider, a travel medicine specialist or one of the websites in annex 1 prior to arranging for any travel outside their home country, since the risk of specific infectious diseases varies from time to time in every country. This is particularly important when seafarers may eat ashore, spend long periods ashore during an extended port call, or spend time ashore either prior to joining or after leaving their ship, but these situations are outside the scope of this document. The advice given here refers to those working on board with only brief periods of shore leave in port.

3.2 Within Europe, USA and Canada, Japan, Australia and New Zealand the risks are usually no greater than those onshore in the UK and no immunisations over and above those given to the general population in the UK are recommended (see annex 2). It is important in any case to ensure that these general immunisations are in date. This situation is of course subject to change over time.

3.3 For all other parts of the world, location specific requirements must be determined and compared to the immunisations that the seafarer has received. A wider range of immunisations is likely to be needed and these must be carried out before the voyage and kept up to date as long as they are relevant. A full course of immunisations may take a month to complete and take effect so prior planning is essential.

3.4 In tropical and subtropical regions, malaria and other insect borne diseases are a particular risk. Insect avoidance and malaria prevention and treatment should be considered well before travel (see annex 3).

4. Job related risks

4.1 An informed clinical risk assessment is required with respect to vaccination for some diseases in some or all seafarers, considering the nationality and origin of both crew and passengers and the route of the vessel. Stringent hygiene measures should remain the mainstay of prevention of infection, however;

4.2 Maintenance of sewage systems - seafarers whose work involves maintaining shipboard sewage systems are at an increased risk of faecal contamination. Immunisation against hepatitis A is recommended for workers at risk of repeated exposure to raw sewage following a local risk assessment. In areas where typhoid is endemic, vaccination for typhoid should also be considered.

4.3 Food handlers - these seafarers will, if infected, pose a risk of transmitting foodborne infections to other crew members or passengers. Although immunisation for food handlers is not routine in the UK, seafarers should have typhoid and hepatitis A vaccinations if they originate from or are working in an area where these infections are likely.

4.4 Cruise liners - where large numbers of susceptible passengers are carried, as on cruise ships, outbreaks of respiratory disease such as Covid-19 and influenza can be severe in terms of time off work and risk to passengers. Crew immunisation should be considered.

4.5 Work on deck - crew who are required to spend periods on deck in places where malaria and/or dengue is common will be at high risk of mosquito bites, and bite avoidance (see annex 3) with correct use of antimalarial tablets where appropriate is particularly important. This is a [map showing where malaria is present](https://www.who.int/data/gho/data/themes/malaria)
(<https://www.who.int/data/gho/data/themes/malaria>)

4.6 Travel away from vessel - If a seafarer is planning to spend time away from the ship and harbour area either on business or for leisure, additional advice is needed before departure on whether the precautions used on account of the ship's voyage remain adequate.

4.7 Seafarers providing medical care - health professionals working on board ship should be fully immunised, including confirmed protection against hepatitis B. The provision of routine hepatitis B immunisation may need to be considered if non-health care specialists are regularly required to provide such care.

5. Responsibilities

5.1 Shipowners and employers have a duty to protect the health and safety of workers (so far as reasonably practicable). This general duty of care includes minimising the risks of infection considering the routes and destinations of ships by:

- making arrangements for infection control on board, ensuring that hygiene measures are effective and suitable
- making seafarers aware of the risks of infection
- supplying medical stores with these risks in mind including where necessary malaria supplies and protection from insect bites
- providing and paying for relevant pre travel preventive measures such as immunisations and malaria prophylaxis.

The discharge of these duties needs to be encompassed within a management system which is effective, and which can be audited.

5.2 Before a seafarer is appointed to a ship, the maritime employer, including the manning agency or ship management company where they formally employ the seafarer(s), must as far as reasonably practicable establish where each crew member will be travelling to and the risks of infection. They should then –

- check the immunisation status of the seafarer
- ensure that any missing immunisations are given
- ensure that preventative medication for malaria has been started where appropriate.
- check that if required appropriate malaria medications and insect bite avoidance measures (e.g. insect repellent sprays) are available on board
- if the destinations are not known, a risk assessment should be carried out by the shipowner who then has the duty to arrange these measures even when they are not the employer of the seafarer. The same applies if the voyage pattern is changed.

5.3 Once full information regarding the itinerary of the proposed vessel is available, travel advice, relevant immunisations and malaria prophylaxis, as well as medical reasons for not giving one of the specific antimalarial treatments may be provided (for a fee) by:

- some but not all doctors approved by the MCA to undertake seafarer medical examinations (approved doctors). Note that such advice is not part of the statutory ENG1 medical but may be provided as an additional service.

- specialist travel health clinics
- company OH providers

In the UK, routine immunisations recommended for all members of the population (see annex 2) are provided by General Practitioners, usually under the NHS.

5.4 It is the seafarer's responsibility to comply with infection control measures instigated by the employer. The seafarer should ask for and keep securely a record of immunisations administered and any specific individual requirements concerning the use of malaria medication including possible side effects and present it as requested by their employers or health care personnel. They should also be aware of the importance of informing any doctor of their recent locations of work if they develop symptoms such as a flu-like illness, fever, or diarrhoea, especially if they have recently travelled from a vessel in an area subject to specific infections. The seafarer's record should be checked before departure by the shipowner and finally by the ship's master.

6. Schedules for immunisation

6.1 The information in annex 2 to this notice indicates the requirements for immunisation in broad terms. The health care professional providing the immunisations will be able to check precise recommendations and advise on the correct immunisation schedule

7. More information

Seafarer Safety and Health Branch Maritime and Coastguard Agency
Bay 2/17
Spring Place
105 Commercial Road
Southampton
SO15 1EG

Telephone: +44 (0)203 81 72835

Email: medical@mcga.gov.uk

Website: www.gov.uk/mca (<https://www.gov.uk/mca>)

General enquiries: info@mcga.gov.uk

Please note that all addresses and telephone numbers are correct at time of publishing.

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Annex 1 Useful references for further information

The Code of Safe Working Practices for Merchant Seafarers

(https://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/938726/Code_of_Safe_Working_Practices_for_Merchant_Seafarers_Amendment_5_Oct_2020_v.2.pdf). Also available from www.tsoshop.co.uk (<http://www.tsoshop.co.uk/>) or Tel: 0333 202 5070

The Ship Captain's Medical Guide – 23rd Edition. Maritime and Coastguard Agency published 2019 by TSO (as above) should be carried on board – available as above.

Wellbeing at Sea: A Pocket Guide for Seafarers published 2020 by TSO (as above)

Travel Health advice

- Foreign and Commonwealth Office, General travel health and safety www.fco.gov.uk/travel/ (<http://www.fco.gov.uk/travel/>)
- National Travel Health Network and Centre (NaTHNaC), Travel Health Pro section <https://travelhealthpro.org.uk/> (<https://travelhealthpro.org.uk/>)
- The World Health Organization: International Travel and Health; Vaccination Requirements and Health <https://www.who.int/travel-advice> (<https://www.who.int/travel-advice>)
- “Fit for Travel” is a public access website provided by the NHS which gives travel health information for people travelling abroad from the UK www.fitfortravel.scot.nhs.uk (<http://www.fitfortravel.scot.nhs.uk/>)

- The International Society of Travel Medicine: an information resource for both travel medicine practitioners and travellers www.istm.org (<http://www.istm.org/>)
- Royal College of Nursing Travel Health advice <https://www.rcn.org.uk/clinical-topics/public-health/travel-health> (<https://www.rcn.org.uk/clinical-topics/public-health/travel-health>)
- World Health Organization International Health Regulations (2005)

ISBN-13 9789241580410. International health regulations (who.int) (https://www.who.int/health-topics/international-health-regulations#tab=tab_3)
Cruise ships

- CDC Travellers' Health. Cruise Ship Travel: Health recommendations. <https://wwwnc.cdc.gov/travel/page/cruise-ship> (<https://wwwnc.cdc.gov/travel/page/cruise-ship>)
- CDC. Vessel Sanitation Program. <http://www.cdc.gov/nceh/vsp> (<http://www.cdc.gov/nceh/vsp>)

Immunisations

- Department of Health, Immunisation against infectious disease - 'The Green Book' (<https://www.gov.uk/government/collections/immunisation-against-infectious-disease-the-green-book>).
- For the UK Immunisation schedule see Chapter 11 (https://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/1060682/Greenbook-chapter-11-11Mar22.pdf)

Medications and vaccines

- MedicinesComplete, from the Royal Pharmaceutical Society includes the British National Formulary. https://about.medicinescomplete.com (<https://about.medicinescomplete.com/>)

Annex 2 UK Routine and job specific immunisations and geographical requirements

1. UK Routine Immunisations

1.1 In the UK children are routinely vaccinated against the following infections. It should be borne in mind however that older seafarers or those born or brought up in different countries may have been vaccinated differently.

- Diphtheria
- Haemophilus influenzae type b (Hib)
- Hepatitis B
- Human papillomavirus (certain serotypes)
- Measles
- Meningococcal disease (certain serogroups)
- Mumps
- Pertussis (whooping cough)
- Pneumococcal disease (certain serotypes)
- Polio
- Rotavirus
- Rubella
- Tetanus

1.2 For more detail and up to date information on both childhood and adult immunisation schedules see the UK publication Immunisation against infectious disease (<https://www.gov.uk/government/collections/immunisation-against-infectious-disease-the-green-book>).

2. Additional immunisations

2.1 Apart from identifying which childhood vaccinations have been given and ensuring that these are up to date, employers or shipowners may wish to consider the advantages of vaccinating crew against other illnesses (e.g. chicken pox, influenza, Covid-19) in view of the risk to adults in contracting these diseases, impact on safe manning in the event of an outbreak on board and costs or delays incurred due to port health restrictions.

2.2 Further immunisations should be considered for travel outside of NW Europe, N America, Japan, Australia & New Zealand, or in consideration of local endemic or epidemic disease, including but not limited to:

- Hepatitis A
- Hepatitis B
- Typhoid
- Meningococcal meningitis (other serotypes)

2.3 Yellow fever vaccination is required for entry into many countries when arriving from countries with risk of yellow fever transmission (sub-Saharan Africa, parts of Central and Southern America). Port Authorities may require an internationally recognised certificate.

2.4 Job related risks are discussed in section 4 of the MGN above.

Annex 3 Insect borne diseases including malaria

1. Insect borne diseases

1.1 A number of infections such as malaria, dengue, zika virus and yellow fever are transmitted by mosquitoes, although ticks and other insect or animal vectors may be encountered ashore.

1.2 Anyone who falls ill whilst in or after leaving an area with risk of these conditions should inform a doctor or other appropriate personnel immediately that they have been exposed to risk of insect borne infection. This applies up to twelve months after return from a malarial area. It is particularly important to seek urgent diagnosis and treatment for any feverish illness even when preventative treatment has been taken.

1.3 Avoidance of insect bites should be the mainstay of prevention of insect borne disease in tropical areas. Although specific preventative measures are available for malaria and yellow fever, this is not the case for other infections, and failure of treatment is a possibility.

- wear long-sleeved shirts and long trousers when going on deck or ashore
- use insect repellent on any exposed skin
- keep openings (e.g. doors, portholes, ventilators and hatches) closed where possible
- use mosquito wire-screening and nets on any openings
- use anti-mosquito preparations or insecticides, treating compartments with insecticide spray and killing any mosquitoes found
- use filtered air conditioning where available
- if seafarers need to stay ashore in non-air-conditioned accommodation, beds should have nets. Nets should preferably be impregnated with permethrin, and well tucked in with no tears or holes

2. Malaria Prophylaxis (prevention)

2.1 Since immunisation against malaria is not available, a course of preventive medication is required when entering many tropical areas.

2.2 Detailed information on risk is needed to decide whether or not medication is required in a specific area, and what medication to use. The shipowner will need to obtain advice from a travel medicine specialist on the particular regime to be followed for each area. Considerations will include cost, efficacy, local resistance patterns, simplicity of dosage schedule and possible adverse effects of the medication.

2.3 There are reliable Internet sources, which specify the current malaria risk in different locations, as well as providing information on any drug resistance, listed in Annex 1 to this Notice. Additional valid local knowledge may be relevant to particular ports. Specific information is not provided here, as recommended regimens are subject to change.

2.4 It is the shipowner's responsibility to ensure that a vessel entering malarious areas carries appropriate medication to prevent malaria in sufficient quantities and to ensure that ships' masters are familiar with the advice given with the medication.

2.5 Medication should be started from days to weeks in advance of arriving in the affected area depending on the medication used, so forward planning is essential. Medication must also continue for the specified time after leaving the area. It is the master's responsibility to determine when anti-malarial tablets should be started and finished, based on the timing of 1. visits to areas where there is a risk of infection.

2.6 Compliance is essential, as most deaths occur in those who do not take the tablets for the required period or in the recommended dosages.

2.7 Some treatments are suitable for all seafarers, but others require the individual to have medical clearance before administration. This is important where drug-resistant forms of malaria exist, and as a result, the first choice medications cannot be used. Doses and choice of medicine may have to be altered when someone has another illness or is taking treatment for another medical condition. The Travel clinic, OH adviser or Approved Doctor will be able to advise the seafarer whether they have any medical reasons for not using particular antimalarial medicines.

2.8 Insect bite avoidance measures remain important regardless of any medication taken since even where appropriate medication has been taken the risk of contracting malaria remains.

2.9 Advice for seafarers who remain on board or are only going as far as the dockside or adjacent areas is not the same as that for travellers to rural areas, as risk in many (but not all) ports is much lower than inland. Crew joining or leaving ships in malarious areas may therefore need prophylactic treatment when those who are already on board may not.

3. Standby treatment

3.1 Appropriate medication for treatment of malaria should be carried to be used in the event of a suspected case. Such medication should only be given if clinically indicated.

3.2 The agent used for the treatment of malaria should be different from the drugs used for prevention, both to minimise drug toxicity and due to concerns over drug resistance.

3.3 Based on specialist advice, the carriage of malaria treatment alone may be justified where the risks are low.

3.4 Treatment must always be accompanied by contact with a tele medical advisory service for recommendations on subsequent action including the

resumption of malaria prophylaxis. Anyone who has been treated for malaria or suspected malaria must have a medical examination at the next port of call. Serious cases need urgent hospitalisation.

3.5 More detailed guidance on diagnosis and care can be found in The Ship Captain's Medical Guide.



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